

## Chapter 6, Problem 43P

### Problem

The current in a 150-mH inductor increases from 0 to 60 mA (steady state). How much energy is stored in the inductor?

### Step-by-step solution

#### Step 1 of 2

Energy stored in the inductor is,

$$w_L = \frac{1}{2} L i^2 \Big|_0^{60 \text{ mA}}$$

#### Step 2 of 2

Substitute 150 mH for  $L$  and 0 to 60 mA for  $i$ .

$$w_L = \frac{1}{2} \times 150 \times 10^{-3} \times (60 \times 10^{-3})^2 - 0$$

$$w_L = 270 \text{ } \mu\text{J}$$

Energy stored in 150 mH inductor is 270  $\mu\text{J}$ .